

Real-Time Off-Tracking Prediction and Swept-Area Collision Warning for Articulated Trucks: From MATLAB Physical Modeling to Embedded Implementation

GuanJien

Abstract—Articulated heavy trucks exhibit significant low-speed off-tracking during turning maneuvers, leading to hazardous inward intrusion of the trailer path. This paper presents a complete geometric and embedded framework starting from MATLAB kinematic modeling to real-time implementation on ESP32-S3. A single-unit model is first established with full-horizon PID control and instantaneous center of curvature (ICC) based off-tracking analysis. The formulation is then extended to an articulated tractor-trailer configuration with a geometrically rigorous TF off-tracking metric derived from trailer instantaneous curvature center (ICCT) intersection. A right-side swept-area danger zone is constructed as a union of boundary tiles. Based on these principles, a short-horizon open-loop prediction algorithm is deployed on embedded hardware and fused with mmWave radar detection for spatial risk evaluation. The proposed method converts geometric off-tracking into a spatial occupancy prediction problem that is lightweight, interpretable, and suitable for real-time deployment.

Index Terms—articulated vehicle, off-tracking, swept area, kinematic prediction, embedded system, mmWave radar

I. INTRODUCTION

Articulated trucks frequently cause right-turn accidents due to severe geometric off-tracking. During low-speed turning, the trailer rear axle follows a smaller radius than the tractor front axle, producing inward intrusion. Existing blind-spot warning systems rely on static distance thresholds and fail to account for vehicle geometry evolution.

This work proposes a complete pipeline:

- MATLAB physical modeling
- Geometric off-tracking quantification
- Swept-area spatial formulation
- Embedded short-horizon prediction
- Radar-based spatial matching

II. MATLAB SIMULATION I: SINGLE-UNIT VEHICLE PHYSICAL MODEL

A. Modeling Assumptions

The vehicle is modeled under:

- Planar motion
- Pure rolling (no lateral slip)
- Constant wheelbase l
- Reference points: front axle A , rear axle B

B. Heading Definition

$$\theta(t) = \text{atan2}(y_A - y_B, x_A - x_B) \quad (1)$$

This defines orientation as the four-quadrant inverse tangent of the axle vector.

C. Kinematic Equations

Rear axle speed:

$$v_B = v_A \cos \alpha \quad (2)$$

Yaw rate:

$$\dot{\theta} = \frac{v_A \sin \alpha}{l} \quad (3)$$

Discrete integration:

$$x_{B,k+1} = x_{B,k} + v_B \cos \theta_k \Delta t \quad (4)$$

$$y_{B,k+1} = y_{B,k} + v_B \sin \theta_k \Delta t \quad (5)$$

D. Wheelbase Constraint Enforcement

$$\tilde{l} = \|A - B\| \quad (6)$$

If $\tilde{l} \neq l$, correction is applied along $\mathbf{AB}$ direction to enforce strict geometry.

E. PID Steering Control

$$e = \theta_{ref} - \theta \quad (7)$$

$$\alpha = K_p e + K_i \int e \, dt + K_d \frac{de}{dt} \quad (8)$$

With saturation:

$$|\alpha| \leq \alpha_{max} \quad (9)$$

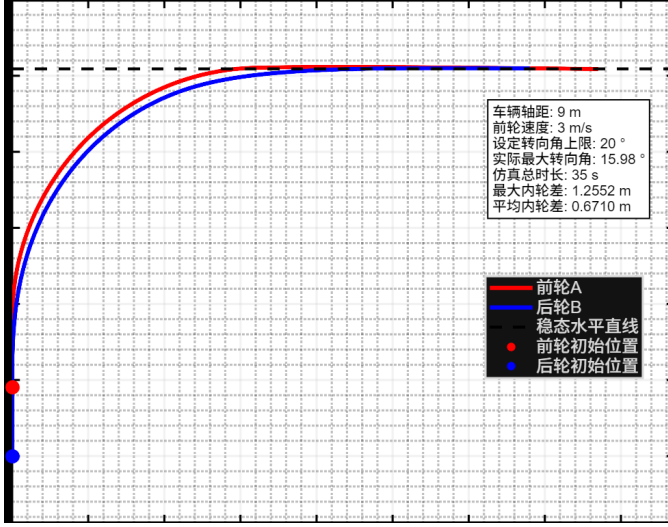


Fig. 1. Single-unit vehicle trajectory.

F. ICC-Based Off-Tracking

Velocity normals:

$$\mathbf{n}_A = (-v_{Ay}, v_{Ax}) \quad (10)$$

$$\mathbf{n}_B = (-v_{By}, v_{Bx}) \quad (11)$$

Intersection yields ICC O .

$$d_{inner} = R_A - R_B \quad (12)$$

III. MATLAB SIMULATION II: ARTICULATED TRACTOR-TRAILER MODEL

A. Geometric Definitions

$$\theta_t = \theta - \phi \quad (13)$$

$$x_H = x_B + l_h \cos \theta \quad (14)$$

$$x_T = x_H - L \cos \theta_t \quad (15)$$

This strictly enforces $|H - T| = L$.

B. Articulation Dynamics

$$\dot{\phi} = \frac{v \sin \alpha}{l} - \frac{v \cos \alpha \sin \phi}{L} \quad (16)$$

Placeholder: TF geometric definition

Fig. 2. TF off-tracking geometry

C. TF Off-Tracking Metric

Trailer instantaneous curvature:

$$R_t = \frac{\|v_T\|}{\omega_t} \quad (17)$$

Left normal:

$$\mathbf{n}_L = (-\sin \theta_t, \cos \theta_t) \quad (18)$$

ICCT:

$$ICCT = T + R_t \mathbf{n}_L \quad (19)$$

Line intersection:

$$\ell(t) = T + t(ICCT - T) \quad (20)$$

TF defined:

$$TF = \|T - F\| \quad (21)$$

IV. SWEEPED-AREA DANGER ZONE CONSTRUCTION

Right normal:

$$\mathbf{n}_R = (\sin \theta_t, -\cos \theta_t) \quad (22)$$

$$H_r = H + \frac{w}{2} \mathbf{n}_R \quad (23)$$

$$T_r = T + \frac{w}{2} \mathbf{n}_R \quad (24)$$

Tile:

$$Q_k = [H_r(k), T_r(k), T_r(k+1), H_r(k+1)] \quad (25)$$

Union:

$$\mathcal{S} = \bigcup Q_k \quad (26)$$

V. EMBEDDED REAL-TIME PREDICTION ON ESP32

Closed-loop removed; measured inputs:

$$v, \quad \alpha, \quad \phi$$

Prediction:

$$\theta_{k+1} = \theta_k + \frac{v \sin \alpha}{l} \Delta t \quad (27)$$

$$\phi_{k+1} = \phi_k + \dot{\phi}_k \Delta t \quad (28)$$

Horizon:

$$T_p = 1 \text{ s}$$

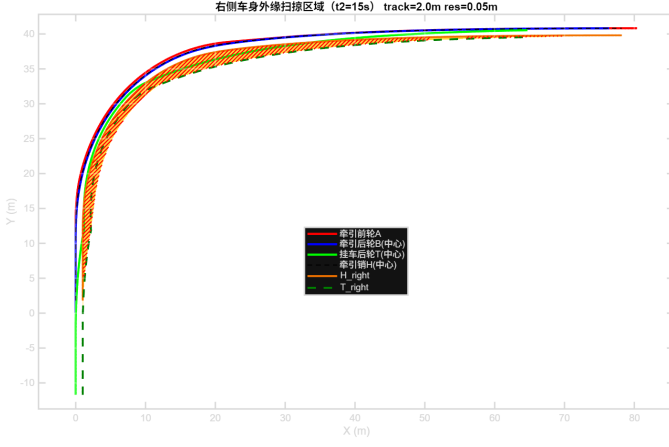


Fig. 3. Swept danger zone.

VI. RADAR SPATIAL MATCHING

Radar detection:

$$x_i = r_i \cos \beta_i \quad (29)$$

$$y_i = r_i \sin \beta_i \quad (30)$$

Warning:

$$\exists i, k : (x_i, y_i) \in Q_k \quad (31)$$

VII. EMBEDDED SOFTWARE ARCHITECTURE

Modules:

- Config
- Sensor Interface
- Kinematics
- Sweep Prediction
- Radar Parser
- Geometry Engine
- Alarm Controller

Loop frequency: 50 Hz.

Worst-case execution ≤ 5 ms.

VIII. COMPUTATIONAL COMPLEXITY

Prediction:

$$O(N_p)$$

Matching:

$$O(N_p \cdot N_{radar})$$

Feasible for ESP32-S3.

IX. EXPERIMENTAL VALIDATION

- MATLAB vs Embedded trajectory comparison
- Static obstacle intrusion
- Dynamic pedestrian intrusion
- False positive/negative rate

X. DISCUSSION

Advantages:

- Physics-consistent
- Interpretable geometry
- Lightweight

Limitations:

- Low-speed assumption
- Constant-input horizon

XI. CONCLUSION

This paper presents a complete pipeline from MATLAB physical modeling to embedded swept-area warning. The transformation of off-tracking into spatial occupancy prediction enables practical real-time deployment.